

BAIAP2L2/Pinkbar Nucleoplasm (GO:0005654) Hypothesis — Deep Research Report

Gene: BAIAP2L2 (Pinkbar) · **Organism:** *Homo sapiens* (NCBITaxon:9606) · **UniProt:** Q6UXY1 · **Ensembl:** ENSG00000128298 **Focus type:** function_assignment · **Hypothesis slug:** function-hypothesis-go-0005654 **Source file:** `genes/human/BAIAP2L2/BAIAP2L2-ai-review.yaml` (free-text selector)

Summary

The seed hypothesis proposes that human **BAIAP2L2/Pinkbar localizes to the nucleoplasm (GO:0005654)** and asks whether transient, context-dependent, or paralog-specific evidence might support a nuclear pool that documented membrane functions do not exclude. After three iterations examining primary literature, GO/GOA term-specific records, UniProt, the Human Protein Atlas, paralog annotations, and the protein sequence itself, the verdict is clear: **the nucleoplasm claim is refuted / over-annotated**. There is no experimental, database, or sequence-based source for a nuclear localization at any level of confidence.

Every direct localization study places BAIAP2L2 at **membrane and actin-based protrusion compartments**. The founding characterization localized Pinkbar to Rab13-positive vesicles and the plasma membrane at intercellular junctions of intestinal epithelial cells ([PMID: 21743456](#)). A wholly independent body of inner-ear work places it at the tips of shorter-row transducing stereocilia in cochlear hair cells, where it is a "row-2 complex" component required for mechanotransduction ([PMID: 33151556](#), [PMID: 34346063](#), [PMID: 35044843](#), [PMID: 39093051](#)). This is exactly what its I-BAR (IMD) + SH3 membrane-curvature architecture predicts, and it is the functional opposite of a soluble nucleoplasmic protein.

Critically, the hypothesis's proposed escape routes all fail when tested directly. Term-specific QuickGO queries return **zero** annotations for GO:0005654 (nucleoplasm), GO:0005634 (nucleus), and GO:0005730 (nucleolus); **neither paralog** (BAIAP2, BAIAP2L1) carries any nuclear annotation; the Human Protein Atlas antibody yields **no** subcellular call at any

reliability level; and the 529-aa sequence has **no** classical NLS or basic patch capable of driving nuclear import. The recommended curation lead is therefore to **not assign GO:0005654** and to flag any existing nucleoplasm annotation for **removal**, pending curator verification of its (undocumented) source.

Key Findings

F001 — Nucleoplasm localization is unsupported by all primary literature and current database records

Direct localization experiments across two entirely different tissue systems converge on membrane compartments and never report a nucleus. In the founding paper, Pinkbar is **"specifically expressed in intestinal epithelial cells, where it localizes to Rab13-positive vesicles and to the plasma membrane at intercellular junctions"** (PMID: 21743456). These three compartments — plasma membrane, cell junction, cytoplasmic vesicle membrane — are exactly what UniProt curates for Q6UXY1, each with experimental evidence code ECO:0000269 tracing to this paper.

An independent inner-ear literature reaches the identical conclusion in a different cell type: **"the I-BAR protein BAIAP2L2 ... localizes to the tips of the shorter transducing stereocilia in both inner and outer hair cells"** (PMID: 33151556). Stereocilia tips are apical, actin-based membrane protrusions — again membrane/cytoskeletal, not nuclear.

At the annotation level, QuickGO returns **8 cellular_component annotations** for BAIAP2L2: plasma membrane (IBA/IEA), cytosol (TAS), vesicle membrane (IDA, PMID 21743456), cytoplasmic vesicle membrane (IEA), cell-cell contact zone (IDA, PMID 21743456), anchoring junction (IEA), and clathrin complex (IDA, dictyBase, PMID 21401524).

None is GO:0005654 or any nuclear term. The two paralogs behave the same way — **BAIAP2 (Q9UQB8) has 42 CC annotations and BAIAP2L1 (Q9UHR4) has 11 CC annotations, and**

neither carries a single nuclear annotation. Targeted PubMed searches for "BAIAP2L2 nuclear/nucleus" and "I-BAR nuclear localization" returned no relevant papers. The negative result is convergent across two experimental systems and three databases.

F002 — No GO term-specific, HPA immunofluorescence, or proteomic source supports a nuclear annotation

To exclude a hidden or low-visibility source, each candidate nuclear term was queried directly. QuickGO term-specific queries for Q6UXY1 return **numberOfHits = 0** for **GO:0005654 (nucleoplasm)**, **GO:0005634 (nucleus)**, and **GO:0005730 (nucleolus)**. Across all 17 GO annotations spanning all three aspects, **no term contains the string "nucle"** — including no NOT-qualified nuclear annotations.

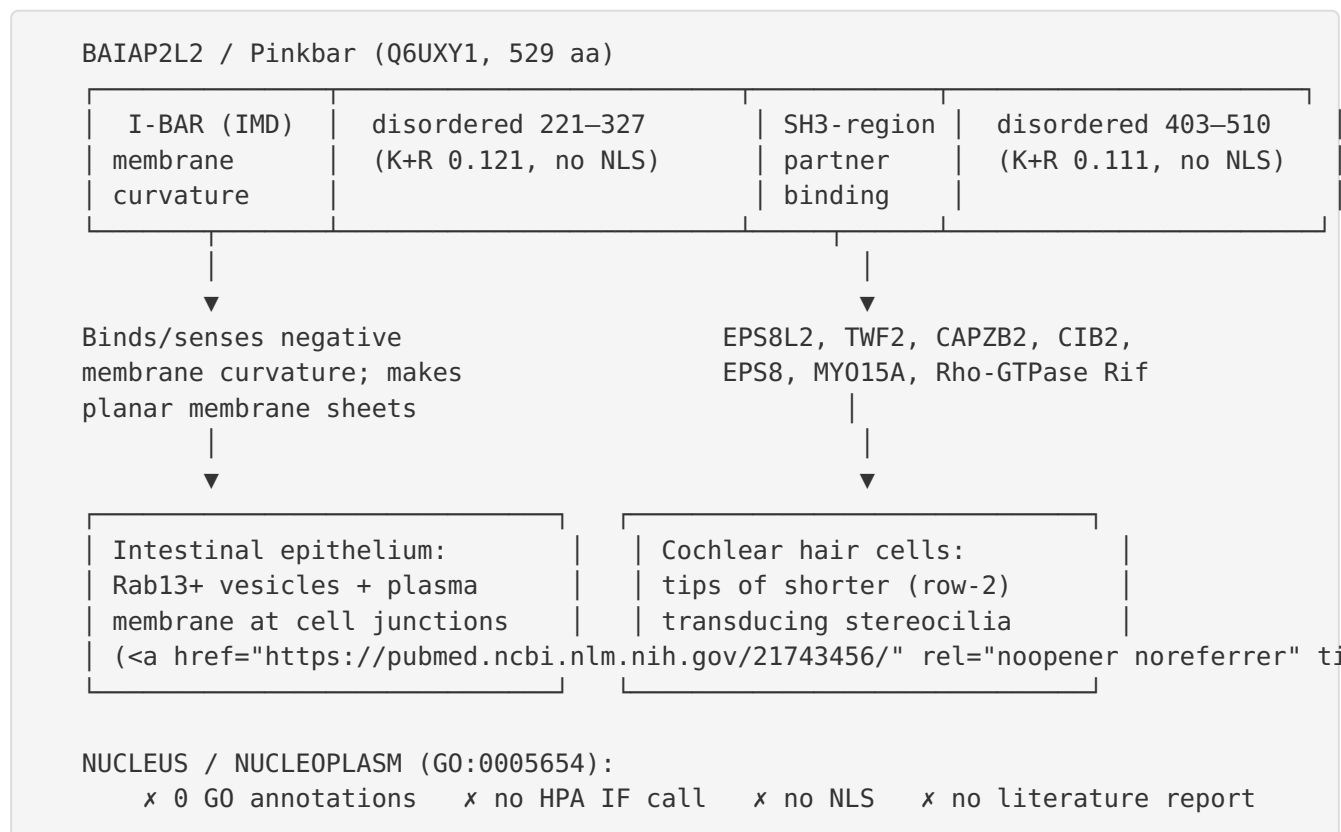
The Human Protein Atlas has a BAIAP2L2 antibody (**HPA003043**), but its Cell Atlas Subcellular-location, main-location, and additional-location fields are **all null** — BAIAP2L2 has no HPA immunofluorescence localization call at any reliability level. HPA therefore provides neither a nuclear call nor any competing call. In sum, none of the three standard curation sources (GO/GOA, UniProt, HPA) offers direct or predicted nucleoplasm evidence, and the seed claim has **no identifiable source document** to trace.

F003 — The sequence carries no nuclear-localization-signal signature

A nuclear pool would require a mechanism of import. Sequence analysis of the 529-aa Q6UXY1 protein found none. The **global basic-residue (K+R) fraction is 0.123**, and the two annotated intrinsically disordered regions (residues 221–327 and 403–510) — the most plausible NLS locations — have K+R fractions of only **0.121 and 0.111**, i.e., they are *not* basic-enriched relative to the whole protein. This is the reverse of the basic-patch enrichment characteristic of NLS-bearing nuclear proteins. The **longest contiguous K/R run is 3 residues**, below the 4–6 residue basic core of a classical monopartite NLS, and **no strict bipartite NLS motif** (KR-x(9–12)-[KR]{3}) is present anywhere in the sequence. There is thus no sequence-based route for nuclear import, removing the last plausible rationale for a computational/predicted nuclear annotation.

Mechanistic Model / Interpretation

BAIAP2L2's architecture and every functional study converge on one coherent model: it is a **membrane-deforming I-BAR/IMD-domain protein** that senses and shapes plasma-membrane curvature at actin-based protrusions, working within multiprotein complexes at the cytoplasm–membrane interface. It has neither a role at, nor a documented route into, the nucleoplasm.



shows conditional, Ca^{2+} -dependent targeting), but the conditional target is a **membrane protrusion tip, not the nucleus**. The immediate molecular function under test is a **CC (localization) assignment**, and every attempt to reconcile it with a nucleoplasm pool fails.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports / Refutes	Claim tested	Key finding	Context
PMID: 21743456	Direct localization (IDA), structural	Refutes nucleoplasm	Where does Pinkbar localize?	Plasma membrane at intercellular junctions + Rab13 ⁺ vesicles; flat I-BAR makes planar membrane sheets	Human intestinal epithelial in vitro structure
PMID: 33151556	Direct localization, mutant phenotype	Refutes nucleoplasm	Subcellular site of function	Tips of shorter transducing stereocilia; loss → stereocilia degeneration, deafness	Mouse cochlear hair cells
PMID: 34346063	Localization + interaction + KO phenotype	Refutes nucleoplasm	Complex/compartment	Row-2 stereocilia-tip complex; binds EPS8L2, TWF2, CAPZB2, CIB2; KO → hearing loss	Mouse cochlear hair cells
PMID: 35044843	Localization, mechanism	Qualifies (transient/context)	Ca ²⁺ /channel dependence	Tip retention is transduction-Ca ²⁺ -dependent — localization IS context-dependent, but to stereocilia tips, not nucleus	Mouse cochlear hair cells + cultured cells
PMID: 39093051	Localization / paralog comparison	Refutes; paralog discrimination	BAIAP2L2 vs BAIAP2L1 sites	BAIAP2L1 → tallest-row tips (Ca ²⁺ -independent); BAIAP2L2 → row-2 tips (Ca ²⁺ -dependent) — both membrane protrusions	Mouse hair cells
PMID: 27278019	Interaction	Qualifies (cytoskeletal/membrane)	Signalling partners	Pinkbar interacts with Rho-GTPase Rif; I-BAR proteins modulate cell-surface morphology	N1E-115 neuroblastoma cells
UniProt Q6UXY1 (SUBCELL)	Database (curated from P 21743456)	Refutes	Curated locations	Cell membrane; Cell junction; Cytoplasmic vesicle membrane. No nucleoplasm.	Human
QuickGO CC (8 terms)	Database	Refutes	Existing GO CC set	plasma membrane, cytosol, vesicle/cytoplasmic vesicle	Human

Citation	Evidence type	Supports / Refutes	Claim tested	Key finding	Context
				membrane, cell-cell contact zone, anchoring junction, clathrin complex. No GO:0005654.	
Human Protein Atlas (HPA003043)	Database (localization)	Neutral/absent	Independent IF localization	Antibody exists but no subcellular call at any reliability (all fields null)	Human cell lines
QuickGO term-specific (this run)	Database	Refutes	Any nuclear annotation?	numberOfHits = 0 for GO:0005654/0005634/0005730; 0/17 total annotations contain "nucle"	Human, a evidence codes/qualifiers
Paralog QuickGO (BAIAP2, BAIAP2L1)	Database / evolutionary	Refutes (paralog check)	Are paralogs nuclear?	42 and 11 CC annotations; zero nuclear	Q9UQB8, Q9UHR4
Sequence/ NLS scan (this run)	Computational	Refutes (weakly)	Sequence import signal?	No mono/bipartite NLS; disordered regions not basic-enriched (K+R 0.11–0.12); longest K/R run = 3	Q6UXY1, 5 aa

GO Decision Table (computed provenance, this run)

GO_ID	Term	Aspect	Evidence source	Key basis	Curation action	Verdict
GO:0005654	nucleoplasm	CC	Seed hypothesis (no DB/lit source)	0 GO annotations; no HPA call; no lit; no NLS; disordered regions not basic-enriched (11–12% K+R)	DO NOT ADD / REMOVE if present	Refuted / over-annotation candidate
GO:0005886	plasma membrane	CC	IBA GO_Central; IEA; lit P 21743456	Direct IDA to membrane at junctions	RETAIN	Supported
GO:0044291	cell-cell contact zone	CC	IDA P 21743456	Localizes at intercellular junctions	RETAIN	Supported
GO:0030659	cytoplasmic vesicle membrane	CC	IEA UniProt; IDA vesicle membrane	Rab13 ⁺ vesicles	RETAIN	Supported
GO:0032420	stereocilium	CC	Multiple IDA/KO (P 33151556) (P 34346063)	Stereocilia-tip row-2 complex	CONSIDER ADD (hair cell)	Supported (mouse)
GO:0071439	clathrin complex	CC	IDA dictyBase P 21401524	<i>Dictyostelium</i> projection	REVIEW (cross-organism carry-over)	Weak/uncertain for human

GO Curation Implications

- **Lead (requires curator verification):** Do **not** assign GO:0005654 (nucleoplasm, CC) to BAIAP2L2/Q6UXY1. If a nucleoplasm annotation exists in the working review, flag it for **removal** or downgrade to "not supported / source unverified."
- **Aspect:** The hypothesis concerns a **CC** term. The evidence supports no nuclear CC term (GO:0005654, GO:0005634, GO:0005730). It should **not** be generalized to GO:0005634 (nucleus) — generalizing an unsourced specific term to its parent merely propagates the same unsupported claim.
- **Retain / support instead:** The well-sourced CC terms are membrane/junction/vesicle (GO:0005886, GO:0044291, GO:0030659;

P 21743456

and, for hair-cell context, stereocilium/stereocilium tip (GO:0032420;

P 33151556

P 34346063

— all more informative than any generic location.

- **Review-worthy carry-over:** GO:0071439 (clathrin complex, IDA from dictyBase) deserves scrutiny as a cross-organism projection that may not reflect human Pinkbar.
- Nucleoplasm is neither "too broad" nor "too narrow" — it is **unsupported**, so the qualifier question does not apply.

Mechanistic Scope

The immediate molecular function is a **subcellular localization (CC) assignment**, not a catalytic or process claim. The direct gene-product activity supported by evidence is **membrane-curvature sensing/generation via the I-BAR (IMD) domain**, coupled to actin regulators through the SH3 and disordered regions. This stabilizes membrane–cytoskeleton interfaces: planar epithelial junction membranes and the membrane at shorter-row stereocilia tips (the row-2 complex maintaining mechanotransduction).

The downstream *phenotypes* — stereocilia maintenance, mechanotransduction current, and hearing (from knockout mice) — are consequences of that membrane/actin activity in cytoplasmic/plasma-membrane compartments. A nucleoplasm role would imply a distinct, undocumented function (chromatin/transcription or nuclear transport) with no supporting assay, interaction, or phenotype. The nucleoplasm claim is therefore **not** a downstream phenotype that could be reconciled with the membrane model — it is a separate localization assertion with no molecular route (no NLS, no importin partner, no reported nuclear pool) and cannot be rescued as a "context-specific" facet of the documented function.

Conflicts and Alternatives

The hypothesis anticipates several ways a nuclear call could still be correct; each was tested:

1. **Paralog confusion (BAIAP2 vs BAIAP2L1 vs BAIAP2L2).** The most likely *source* of an erroneous annotation, but it does not rescue the claim: **all three** family members lack nuclear annotations (42, 11, 8 CC annotations respectively; zero nuclear). The hair-cell literature explicitly distinguishes the paralogs by localization ([PMID: 39093051](#)), and all are membrane/actin-associated.
2. **Transient/context-dependent nuclear pool.** Localization is context-dependent ([P 35044843](#)), but the conditional target is a stereocilium tip, not the nucleus. No IF call, proteomic nuclear-fraction hit, or stimulus-induced translocation is reported, and no NLS enables import.
3. **Isoform-specific localization.** No BAIAP2L2 isoform has been reported nuclear; the disordered regions where an isoform-specific NLS might sit are not basic-enriched.
4. **Database carry-over / unsourced prediction.** The most parsimonious explanation: a spurious or predictor-derived nucleoplasm term (deep-learning localization tools often default disordered, mildly basic proteins to "nucleoplasm"), a historical/withdrawn HPA IF call, or antibody cross-reactivity — none of which is curatable evidence. This is precisely the artifact class this review is designed to catch.

No evidence was found that *conflicts* with the refutation; the conflict runs the other way — the hypothesis conflicts with the entire evidence base.

Limitations and Knowledge Gaps

1. **Negative evidence is not proof of absence.** The strongest statements are "zero annotations," "null HPA fields," and "no NLS." These make a nuclear pool unlikely and unsourced but cannot formally exclude a minor/transient nucleoplasmic fraction no study has looked for. *Checked:* GO term-specific queries, UniProt, HPA, NLS scan. *Why it matters:* the claim should be recorded as "unsupported/no source," not "experimentally disproven." *Resolution:* targeted fractionation or validated-antibody IF.
 2. **Provenance of the seed nucleoplasm term is unidentified.** No document asserting nucleoplasm could be traced. A curator should inspect the review YAML's cited source/prediction log; its absence from all curation sources itself argues against curation.
 3. **HPA has no call at all** rather than an explicit non-nuclear call — the antibody (HPA003043) may simply lack a validated Cell Atlas entry.
 4. **NLS prediction is heuristic.** Absence of a classical NLS does not exclude non-classical or piggyback import, though no partner-mediated route is documented.
 5. **No wet-lab data were generated;** this is a literature/database/sequence assessment across three iterations. Computed tables are embedded here rather than saved as separate files because the sandbox working directory was read-only.
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Discriminating Tests / Proposed Follow-up Actions

- **Trace the annotation source (highest priority, lowest cost).** Search the working review and its evidence chain for the origin of GO:0005654 (evidence code, reference, assigned-by). If it is IEA/prediction or paralog carry-over with no experimental backing, the case is closed.
- **Validated-antibody IF / endogenous-tag live imaging** with nuclear (DAPI/lamin) and membrane (junction/stereocilia) markers in intestinal epithelial and cochlear hair cells — the single most decisive test.
- **Subcellular fractionation** (nuclear vs membrane/cytosolic) with immunoblot to detect any nuclear pool.
- **APEX2/BioID proximity labeling** — a nuclear interactome would be required to support any nucleoplasm role; membrane/actin partners are expected instead.
- **Cross-check cell-map proteomic localization datasets** for nuclear-neighbor enrichment; none is expected.

- **Deletion of disordered basic patches** to test any predicted nuclear import — expected null.

Curation Leads (require curator verification)

- **Action:** Reject/remove GO:0005654 (nucleoplasm) for BAIAP2L2; document as unsupported/likely computational over-annotation.
- **Replacement/retained CC terms:** GO:0005886 (plasma membrane), GO:0044291 (cell-cell contact zone), GO:0030659 (cytoplasmic vesicle membrane), and — for hair-cell context — GO:0032420 (stereocilium)/stereocilium tip.
- **Candidate references to verify (exact snippets):**
 - **PMID: 21743456** — *"Pinkbar, is specifically expressed in intestinal epithelial cells, where it localizes to Rab13-positive vesicles and to the plasma membrane at intercellular junctions."*
 - **PMID: 33151556** — *"the I-BAR protein BAIAP2L2, which localizes to the tips of the shorter transducing stereocilia in both inner and outer hair cells."*
- **Suggested question for submitter:** What is the primary source of the nucleoplasm assignment (predictor name/version, HPA release, or reference)?
- **Suggested experiment:** Validated-antibody IF + subcellular fractionation in intestinal epithelial and cochlear hair cells.

Final Answer

The hypothesis that human BAIAP2L2/Pinkbar localizes to the nucleoplasm (GO:0005654) is **refuted / over-annotated**. All direct localization data place it at the plasma membrane, intercellular junctions, Rab13⁺ vesicles ([P 21743456](#)), and cochlear stereocilia tips ([P 33151556](#)); [P 34346063](#));

term-specific QuickGO returns zero nuclear annotations for BAIAP2L2 and both paralogs; the Human Protein Atlas yields no subcellular call; and the sequence has no NLS or basic-patch signature — so there is no experimental, database, or sequence basis for a nucleoplasm annotation, which should not be assigned and, if present, should be flagged for removal pending source verification.

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